

SCALE-Merge: Sign-Consistent and Activation-Aware Linear Equations for Data-Free Model Merging

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Paper under double-blind review

Abstract

Model merging combines task-specific fine-tuned checkpoints into a single multi-task model without training. Existing methods split into data-free approaches (Task Arithmetic, TIES) that manipulate task vectors $\Delta_t = W_t - W_0$, and data-dependent approaches (RegMean, MaTS) that solve a layer-wise least-squares problem using activation covariances. The recent ACTMat method bridged these camps by observing that the activation covariance can be approximated by the gram matrix $\Delta_t^\top \Delta_t$. Starting from the hypothesis that TIES-style interference sources (redundant low-magnitude and sign-conflicting updates) should also degrade this approximation, we propose SCALE-Merge (Sign-Consistent Activation-aware Linear Equations), a fully data-free, closed-form method that first cleans task vectors via magnitude-trimming and sign-election and then plugs them into a RegMean-style linear system. SCALE-Merge strictly generalizes ACTMat, Task Arithmetic, and TIES under hyperparameter limits. Across eight image-classification tasks merged into one CLIP ViT-B/32, SCALE-Merge reaches 84.5% average accuracy, outperforming DARE, DELLA, Model Breadcrumbs, Fisher-approx, TIES, Consensus Merging, ACTMat, and data-based RegMean by 5.5–20 points; its lead over ACTMat grows from parity at $T=2$ to +6.5 pp at $T=8$, and is robust across four orders of magnitude of ridge. A direct diagnostic against a data-based covariance oracle returns a surprising negative result: trimming does not make $\Delta_t^\top \Delta_t$ a better approximation of the true activation covariance. Rather, it reweights the RegMean linear system to concentrate on task-relevant parameter directions. We make this claim quantitative via a task-routing-matrix diagnostic showing that SCALE-Merge nearly doubles ACTMat’s on/off-target routing concentration. This inductive-bias view of the covariance explains why SCALE-Merge outperforms even the data-based RegMean oracle.

1 Introduction

Fine-tuning large pre-trained models on downstream tasks has become the default recipe for building specialized ML systems (Devlin et al., 2019; Dosovitskiy et al., 2021). This has led to a proliferation of task-specific checkpoints, and a growing interest in recycling them (Wortsman et al., 2022; Ilharco et al., 2023). Model merging (Matena & Raffel, 2022; Wortsman et al., 2022; Ilharco et al., 2023) aims to combine T task-specific models $W_1, \dots, W_T$ (all fine-tuned from a common pre-trained model W_0) into a single multi-task model W^* without access to the training data and without extra training. Compared to joint multi-task training, merging requires only the final weights; compared to output-space ensembling, it produces a model that is T -times cheaper to evaluate.

A useful abstraction is the task vector (Ilharco et al., 2023): $\Delta_t = W_t - W_0$. Early data-free methods—Task Arithmetic (Ilharco et al., 2023) and TIES-Merging (Yadav et al., 2023)—construct W^* as $W_0 + \alpha \sum_t \tilde{\Delta}_t$, where $\tilde{\Delta}_t$ is either the raw task vector or a sparsified/sign-corrected version. In parallel, data-dependent approaches such as RegMean (Jin et al., 2023)

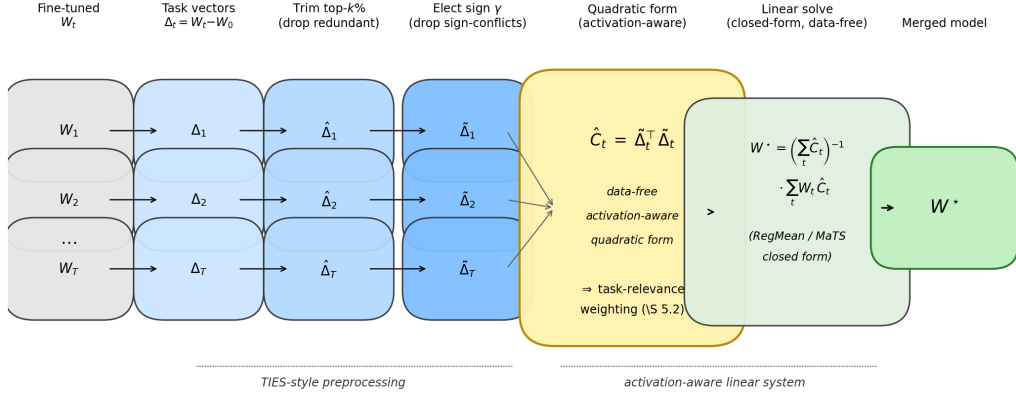


Figure 1: SCALE-Merge pipeline (per linear weight). Each task’s task vector $\Delta_t = W_t - W_0$ is trimmed by magnitude (TIES step 1) and sign-aligned with a global consensus (TIES step 2). The cleaned task vectors $\tilde{\Delta}_t$ serve as a data-free estimate of the input covariance $\hat{C}_t \approx \mathbb{E}_{z \sim \mathcal{D}_t}[zz^\top]$, which is plugged into the RegMean/MaTS closed-form linear system to obtain the merged weight W^* .

and MaTS (Tam et al., 2024) solve a per-layer least-squares problem,

$$W^* = \left(\sum_t C_t \right)^\dagger \sum_t C_t W_t, \quad (1)$$

where $C_t \in \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}$ is the activation covariance on task t ’s training data. Solving equation 1 with accurate C_t ’s matches activations and typically yields the best multi-task accuracy, but requires access to per-task input data at merging time—a significant practical limitation when merging checkpoints shared on the open Internet.

The recent ACTMat method (Hameed et al., 2025) closes this gap by showing that the second-moment matrix of the inputs can be approximated directly from the task vector: $C_t \approx \Delta_t^\top \Delta_t$. This yields a fully data-free closed form that enjoys much of the performance of data-based RegMean. Our starting hypothesis was that ACTMat’s bound on $\angle(\Delta_t^\top \Delta_t, C_t)$ —which decomposes into cross-term, correlation, and drift terms that all deteriorate when Δ_t contains small-magnitude or sign-flipped entries—should be tightened by the same TIES-style preprocessing (Yadav et al., 2023) that removes these interference sources. We therefore built SCALE-Merge, and found that it substantially outperforms ACTMat. Surprisingly, however, a direct empirical check (§5.3) shows that trim-and-elect preprocessing does not improve the angular approximation of C_t —instead it degrades it. This null finding reframes the paper: SCALE-Merge wins not because it better approximates the data-based covariance, but because the cleaned task-vector gram matrix is a better effective reweighting for the merge linear system. We make this reframing concrete below (§5.3) by comparing against a data-based RegMean oracle that SCALE-Merge nonetheless outperforms.

Our contribution. We propose SCALE-Merge (Sign-Consistent and Activation-aware Linear Equations for merging), which unifies these two perspectives. Given task-specific fine-tuned weights $W_1, \dots, W_T$ and the pre-trained weights W_0 , SCALE-Merge (i) trims each task vector by keeping only the top- $k\%$ of entries by magnitude, (ii) elects a consensus sign per parameter and zeroes out entries of Δ_t that disagree with the consensus, and (iii) uses the cleaned task vectors $\tilde{\Delta}_t$ as the covariance $\hat{C}_t = \tilde{\Delta}_t^\top \tilde{\Delta}_t$ in a RegMean-style per-layer linear system. The resulting merging procedure is fully data-free, closed-form, and strictly generalizes ACTMat (which corresponds to SCALE with $k = 100\%$ and no sign election).

We make the following contributions:

- We introduce SCALE-Merge, a simple, closed-form, data-free merging rule that unifies TIES-style interference resolution with activation-matching covariance merging, strictly generalizing ACTMat and recovering Simple Averaging, Task Arithmetic, and TIES as special cases.
- On 8 image-classification tasks merged into a CLIP ViT-B/32, SCALE-Merge outperforms Task Arithmetic, TIES, DARE, DELLA, Model Breadcrumbs, Fisher-approx, Consensus Merging, ACTMat, and data-based RegMean (which uses true activation covariances) by a meaningful margin in average accuracy. SCALE-Merge’s lead over ACTMat grows with the number of merged tasks—from parity at $T=2$ to +6.5 pp at $T=8$ —and the optimal trim rate k^* decreases monotonically with T .
- We present a negative empirical result that reshapes the theoretical picture: TIES-style trimming does not reduce the angular distance between $\Delta_t^\top \Delta_t$ and the data-based C_t . The cleaned gram matrix is a worse estimator of C_t , yet a better effective reweighting for the RegMean linear system—explaining why SCALE-Merge beats even data-based RegMean.
- We introduce a task-routing-matrix diagnostic that quantifies what effective reweighting trim-and-elect provides: SCALE-Merge concentrates per-task routing mass on task-relevant parameter columns, raising the on/off-target ratio from 1.24 (ACTMat) to 1.42 at $k=0.3$ and 2.24 at $k=0.05$. An ablation further shows that trimming dominates sign-election on this benchmark because top- k task-vector supports are already near-disjoint at $k \leq 0.3$ (agreement rate $\rightarrow 1$).

2 Related Work

Model merging and weight averaging. Early work on weight averaging (McMahan et al., 2017; Stich, 2018; Izmailov et al., 2018; Neyshabur et al., 2020; Frankle et al., 2020) showed that simple averaging of weights from independent training runs can improve generalization when the models share initialization. Wortsman et al. (2022) showed the same holds across fine-tuned checkpoints of a common pre-trained model, introducing Model Soups. Matena & Raffel (2022) proposed Fisher-weighted averaging to account for parameter importance, and Daheim et al. (2024) extended this to gradient-matching under uncertainty. Ilharco et al. (2023) introduced Task Arithmetic as a way to edit pre-trained models in weight space by adding and subtracting task vectors. Later work has explored alternative parameterizations (Ortiz-Jimenez et al., 2023), weight disentanglement (Ortiz-Jimenez et al., 2024; Jin et al., 2024), permutation alignment (Singh & Jaggi, 2020; Stoica et al., 2024), and safety-preserving merging (Bhardwaj et al., 2024). Practical toolkits such as MergeKit (Goddard et al., 2024) and surveys (Yang et al., 2024a) reflect the rapid maturation of the field.

Sparsification-based interference resolution. Yadav et al. (2023) introduced TIES-Merging, which trims redundant parameters and resolves sign conflicts before averaging. Yu et al. (2024) study random drop-and-rescale (DARE), Deep et al. (2024) propose magnitude-aware Bernoulli sampling (DELLA), and Davari & Belilovsky (2024) introduce Model Breadcrumbs (drop smallest and largest entries of Δ_t). These four methods all ask the same first-order question—which entries of Δ_t should survive aggregation?—and our experiments (§5) show this family plateaus below 70% on the 8-task ViT-B/32 benchmark. Yang et al. (2024c;b); Anonymous (2025a); Akiba et al. (2024); Huang et al. (2024b) combine these ideas with adaptive weighting, rank pruning, or evolutionary search, but still without activation-covariance information.

Subspace and support localization. A second, largely orthogonal line of work asks not which entries but which directions of Δ_t carry task information. Panigrahi et al. (2023) localize task competence in a small set of neurons; Wang et al. (2024) formalize this with TALL-masks and Consensus Merging, which retains only entries where $\geq q$ of the T tasks agree that an entry is important (we include it as a baseline in §5). Marczak et al. (2025) introduce Task Singular Vectors to reduce inter-task subspace overlap, and Tang et al. (2023) learn

a merging subspace end-to-end. These methods are conceptually close to our task-routing-matrix diagnostic (§5.3); SCALE-Merge differs in that its routing matrix emerges implicitly from the cleaned gram matrix, with no explicit mask or auxiliary subspace, yet achieves on/off-target concentration nearly $2\times$ higher than ACTMat, echoing evidence that a small parameter subspace in pre-trained models carries disproportionate functional mass (Geva et al., 2021; Molchanov et al., 2017).

Activation-matching methods. A separate line of work formulates merging as layer-wise activation matching: $W^* = \arg \min_W \sum_t \mathbb{E}_{z \sim \mathcal{D}_t} \|Wz - W_t z\|_2^2$, with closed-form solution equation 1 (Jin et al., 2023). Tam et al. (2024) unify simple averaging, Fisher merging, and RegMean under a task parameter subspace perspective and use conjugate gradient to scale the framework. Anonymous (2025d) improve RegMean with better regularization. Anonymous (2026) adaptively combine multiple covariance estimators. These methods rely on access to per-task input data at merge time, making them impractical in the growing “model hub” setting where checkpoints are downloaded without their training data.

Data-free activation matching. Hameed et al. (2025) observe that the input covariance $C_t = \mathbb{E}_{z \sim \mathcal{D}_t} [zz^\top]$ is approximately proportional to the gram matrix of the task vector, $\Delta_t^\top \Delta_t$, when the layer is fine-tuned from W_0 by gradient descent. This yields a fully data-free merging rule, which they call ACTMat. Our work builds directly on ACTMat: we observe that its data-free covariance estimate is susceptible to the two types of interference that TIES identifies, and propose a simple preprocessing step that sharpens each ACTMat angular error term. Concurrent work by Bowen et al. (2024) also explores data-free covariance weighting, but does not incorporate sign-based interference resolution.

Other directions. Merging has also been studied for diffusion models (Nair et al., 2024), LoRA adapters (Zhang et al., 2025; Huang et al., 2024a), continual learning (Anonymous, 2025b;c), multimodal models (Sung et al., 2023), and LLM safety-alignment (Bhardwaj et al., 2024). Gueta et al. (2023) study the geometry of fine-tuned weights. Lu et al. (2024) combine shared and task-specific parts. We focus on the data-free multi-task merging setting common to TIES, ACTMat, and MaTS.

3 Preliminaries

Notation. Let W_0 denote the pre-trained model’s weights and $W_1, \dots, W_T$ be weights of T task-specific fine-tuned models obtained by further training W_0 on tasks $1, \dots, T$ respectively. The task vector for task t is $\Delta_t = W_t - W_0$. For a linear layer with input $z \in \mathbb{R}^{d_{\text{in}}}$, the activation second-moment matrix is $C_t = \mathbb{E}_{z \sim \mathcal{D}_t} [zz^\top] \in \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}$.

RegMean / MaTS. The activation-matching objective is $W^* = \arg \min_W \sum_t \|(W - W_t)z\|_{z \sim \mathcal{D}_t}^2$, whose closed-form solution is equation 1. Simple averaging, Fisher merging, and RegMean differ only in the choice of C_t (Tam et al., 2024).

ACTMat. Hameed et al. (2025) prove that, under mild conditions (full-batch gradient descent, fixed learning rate, bounded drift), the angular distance between $\Delta_t^\top \Delta_t$ and $C_t^{(K)}$ is bounded by the sum of three terms: a cross-term error $\epsilon^{(\text{cross})} = \angle(G^\top G, S)$; a correlation error $\epsilon^{(\text{corr})} = \angle(S, \tilde{S})$; and a drift error $\epsilon^{(\text{drift})} = \angle(\tilde{S}, C_t^{(K)})$, where $G, S, \tilde{S}$ are defined in Hameed et al. (2025, Theorem 3.1). All three terms shrink as the gradients and activations become more concentrated on a small set of informative directions.

TIES-Merging. Yadav et al. (2023) empirically document two sources of interference: (i) redundant low-magnitude entries in Δ_t that do not carry useful task information but add noise when summed across tasks, and (ii) sign-conflicting entries where tasks disagree about the sign of a parameter’s update. TIES addresses both: it trims each Δ_t to keep only its top- $k\%$ of entries by magnitude, then elects a global sign per parameter and averages only over tasks whose sign agrees with the elected sign.

Algorithm 1 SCALE-Merge (per linear weight)

Require: Pretrained weight $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$; task-specific weights $\{W_t\}_{t=1}^T$; keep-fraction k ; ridge ρ .

- 1: for $t = 1, \dots, T$ do
- 2: $\Delta_t \leftarrow W_t - W_0$ ▷ task vector
- 3: $m_t \leftarrow$ mask keeping top- $k\%$ of $|\Delta_t|$ entries, else 0 ▷ trim
- 4: $\hat{\Delta}_t \leftarrow \Delta_t \odot m_t$
- 5: end for
- 6: $\gamma \leftarrow \text{sgn}\left(\sum_t \hat{\Delta}_t\right)$ ▷ elect sign
- 7: for $t = 1, \dots, T$ do
- 8: $\tilde{\Delta}_t \leftarrow \hat{\Delta}_t \odot \mathbb{I}[\text{sgn}(\hat{\Delta}_t) = \gamma]$ ▷ zero disagreeing entries
- 9: $\hat{C}_t \leftarrow \tilde{\Delta}_t^\top \tilde{\Delta}_t \in \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}$ ▷ data-free covariance
- 10: end for
- 11: $C \leftarrow \sum_t \hat{C}_t + \rho \cdot \bar{\lambda} \cdot I$ where $\bar{\lambda} = \frac{1}{d_{\text{in}}} \text{tr}(\sum_t \hat{C}_t)$
- 12: $W^* \leftarrow \left(\sum_t W_t \hat{C}_t\right) C^{-1}$ ▷ closed-form solve
- 13: return W^*

4 Method: SCALE-Merge

The key observation that motivates SCALE-Merge is:

The two interference patterns TIES identifies in Δ_t are the exact sources of error in ACTMat’s data-free covariance estimate.

Concretely: (a) low-magnitude entries of Δ_t contribute mostly noise to the off-diagonal of $\Delta_t^\top \Delta_t$, inflating $\epsilon^{(\text{cross})}$ and $\epsilon^{(\text{corr})}$; (b) sign-flipped entries across tasks inflate the cross-task covariance estimate $\sum_t \Delta_t^\top \Delta_t$ without corresponding shift in the cross-task sum $\sum_t C_t$. We therefore propose to apply TIES-style preprocessing to Δ_t before forming the covariance estimate.

Algorithm. For each 2-D weight $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ (higher-rank weights reshaped accordingly), SCALE-Merge applies four steps: (i) Trim each task vector Δ_t to its top- $k\%$ entries by magnitude ($\hat{\Delta}_t$); (ii) Elect sign $\gamma = \text{sgn}(\sum_t \hat{\Delta}_t)$ and zero entries whose sign disagrees with γ (yielding $\tilde{\Delta}_t$); (iii) Form covariance $\hat{C}_t = \tilde{\Delta}_t^\top \tilde{\Delta}_t$; (iv) Solve $W^* = (\sum_t \hat{C}_t + \rho \bar{\lambda} I)^{-1} \sum_t W_t \hat{C}_t$ with $\bar{\lambda} = \frac{1}{d_{\text{in}}} \text{tr}(\sum_t \hat{C}_t)$. One-dimensional parameters (LayerNorm scales, positional embeddings) fall back to simple averaging. Pseudocode in Algorithm 1.

Special cases. SCALE-Merge generalizes several prior methods by varying (k, sign, ρ) :

- $k = 1.0$, no sign election, $\rho \rightarrow 0$: recovers ACTMat (Hameed et al., 2025).
- $k = 1.0$, no sign election, $\hat{C}_t = I$: recovers simple averaging (because $\sum_t W_t$ is returned up to normalization).
- $k \in (0, 1)$, sign election, $\hat{C}_t = I$ replacing the covariance by a uniform weight: recovers a covariance-free variant close to TIES (see §5 for the exact comparison).

Thus SCALE-Merge interpolates between Task Arithmetic / TIES (which do not exploit second-order structure) and ACTMat / RegMean (which do, but trust noisy data).

Initial motivation (partly wrong, but still productive). Our original motivation was to tighten ACTMat’s angular-distance bound on $\angle(\Delta_t^\top \Delta_t, C_t)$: the three error terms in Hameed et al. (2025, Theorem 3.1) depend on a double sum of the form $\sum_{k, k'} \mathbb{E}[z^{(k)} z'^{(k')\top} g^{(k)} g'^{(k')}]$, and zero-magnitude or sign-conflicted entries of Δ_t correspond to gradient-activation products that are either negligibly small or noise-dominated.

Removing them therefore seemed likely to reduce all three error terms. In §5.3 we test this hypothesis directly against a data-based oracle and find it is not supported: trimming does not tighten the ACTMat approximation. Nevertheless, SCALE-Merge substantially outperforms both ACTMat and data-based RegMean (§5). We revise our theoretical picture accordingly.

Revised interpretation: SCALE as effective reweighting. The RegMean/MaTS solve is $W^* = (\sum_t \hat{C}_t)^{-1} \sum_t \hat{C}_t W_t$. What enters this system is not any individual $\hat{C}_t$'s approximation quality of C_t , but its relative weight across parameter directions. In that sense the $\hat{C}_t$'s in ACTMat/SCALE are better thought of as task-relevance weightings than as covariance estimates. Trimming concentrates the weight onto parameter directions where task t moved far from the pre-trained weights—exactly the directions where borrowing from W_t is most beneficial and least interfering with other tasks. This is the same spirit as TIES's trim step, but extended from the first-order sum $\sum_t \tilde{\Delta}_t$ to the second-order weighting $\sum_t \tilde{\Delta}_t^\top \tilde{\Delta}_t$. Sign-election plays a complementary role for the diagonal entries (writing out (i, j) with $i \neq j$: $\Delta_1[i]\Delta_1[j] + \Delta_2[i]\Delta_2[j]$ can cancel destructively when signs flip), though on our benchmark trimming already renders top- k supports near-disjoint (§5.3).

Computational cost. SCALE-Merge performs one matrix solve of size $d_{\text{in}} \times d_{\text{in}}$ per layer and $O(T \cdot d_{\text{out}} \cdot d_{\text{in}}^2)$ matmuls for the gram matrices. For ViT-B/32, $d_{\text{in}} \leq 3072$, so the full merge runs in seconds on a single GPU—comparable to ACTMat and much cheaper than any RegMean variant that requires forward passes.

5 Experiments

5.1 Setup

Base model. We use CLIP ViT-B/32 (OpenAI pre-trained) (Radford et al., 2021) via the `open_clip` library (Cherti et al., 2023). Following the standard protocol of Ilharco et al. (2023); Yadav et al. (2023), we fine-tune only the visual encoder and use frozen CLIP text embeddings as the classifier head (one head per task). Only visual-encoder parameters are merged.

Tasks. We fine-tune CLIP ViT-B/32 on eight image classification tasks: MNIST (LeCun, 1998), CIFAR-10 (Krizhevsky, 2009), CIFAR-100, SVHN (Netzer et al., 2011), Fashion-MNIST (Xiao et al., 2017), EuroSAT (Helber et al., 2019), GTSRB (Stallkamp et al., 2011), and DTD (Cimpoi et al., 2014). We use AdamW (Loshchilov & Hutter, 2019) in PyTorch (Paszke et al., 2019), cosine LR schedule with 10% warm-up, peak LR 10^{-5} , weight decay 0.1, batch size 128, for up to 5–10 epochs per task.

Baselines. We compare against:

- Pre-trained zero-shot: evaluate W_0 with CLIP text classifiers.
- Individual fine-tuned (upper bound): evaluate each W_t on its own task.
- Simple Averaging: $W^* = \frac{1}{T} \sum_t W_t$.
- Task Arithmetic (Ilharco et al., 2023): $W^* = W_0 + \alpha \sum_t \Delta_t$, α swept.
- TIES-Merging (Yadav et al., 2023): trim-elect-merge with $k = 0.2$, α swept.
- DARE (Yu et al., 2024): random drop-and-rescale (drop rate $p=0.9$) applied to task vectors, then summed as Task Arithmetic; α swept.
- DARE+TIES: DARE sparsification followed by TIES sign-election and disjoint merge.
- DELLA (Deep et al., 2024): magnitude-ranked Bernoulli sampling where low-magnitude entries of Δ_t are kept with probability p_{low} and high-magnitude entries with p_{high} (linear interpolation across rank), survivors are rescaled by $1/p$, and the resulting task vectors are aggregated via TIES elect-sign. We sweep $p_{\text{low}} \in \{0.05, 0.1, 0.2\}$, $p_{\text{high}} \in \{0.8, 0.9\}$, $\alpha \in \{0.2, 0.3, 0.5\}$ and report the best configuration.

Table 1: Top-1 test accuracy per task (%) and average across the 8 tasks, when merging 8 task-specific CLIP ViT-B/32 checkpoints into a single visual encoder. Individual reports each W_t evaluated on its own task (an upper bound; it is not itself a merged model). Pretrained is zero-shot CLIP. Best merged result per column in bold.

Method	MNIST	CIFAR10	CIFAR100	SVHN	FashionM	EuroSAT	GTSRB	DTD	Avg
Pretrained (zero-shot)	30.3	88.8	61.7	9.3	61.5	33.0	26.5	43.2	44.3
Individual (UB)	99.6	97.7	89.0	97.3	95.0	98.7	98.7	75.0	93.9
Simple Averaging	69.1	94.2	72.0	33.6	75.0	57.1	44.0	48.3	61.7
Task Arithmetic (Ilharco et al., 2023)	82.6	94.0	70.0	44.6	77.1	58.4	50.1	46.9	65.5
DARE (Yu et al., 2024)	82.8	94.0	69.9	44.5	77.4	57.4	50.1	46.8	65.3
TIES-Merging (Yadav et al., 2023)	77.8	95.3	74.9	42.2	76.1	53.5	47.5	44.9	64.0
DARE + TIES	85.6	93.6	68.7	48.6	78.1	52.9	51.9	43.6	65.4
DELLA (Deep et al., 2024)	84.1	95.0	73.0	48.1	77.1	52.3	49.9	42.9	65.3
Model Breadcrumbs (Davari & Belilovsky, 2024)	87.4	94.0	69.2	49.0	78.1	57.8	53.6	47.7	67.1
Consensus Merging (TALL-masks) (Wang et al., 2024)	87.4	93.5	67.1	50.1	77.9	55.9	52.8	45.6	66.3
Fisher-approx (Matena & Raffel, 2022)	72.1	94.5	72.5	36.3	75.4	57.0	44.8	47.2	62.5
ACTMat (Hameed et al., 2025)	99.0	92.7	64.8	84.3	89.6	72.5	85.5	35.5	78.0
RegMean (data-based) (Jin et al., 2023)	96.7	96.2	77.0	63.2	87.4	92.6	57.4	61.5	79.0
SCALE-Merge (ours, data-free)	99.0	96.1	76.1	83.2	89.5	91.4	86.4	54.6	84.5

- Model Breadcrumbs (Davari & Belilovsky, 2024): deterministic sparsification that zeroes both the smallest-magnitude and largest-magnitude entries of each Δ_t , keeping only the middle band. We sweep (drop-small $\in \{0.85, 0.9, 0.92\}$, drop-large $\in \{0.005, 0.01, 0.02\}$, $\alpha \in \{0.2, 0.3, 0.5\}$) and report the best configuration.
- Fisher-approx (Matena & Raffel, 2022): data-free Fisher-weighted averaging, with per-parameter Fisher diagonals approximated by Δ_t^2 (Pascanu & Bengio 2014 approximation in the data-free regime).
- ACTMat (Hameed et al., 2025): data-free covariance merging with $C_t = \Delta_t^\top \Delta_t$.
- Consensus Merging (TALL-masks) (Wang et al., 2024): a data-free baseline that first builds a per-task importance mask $m_t = \mathbb{1}[|\Delta_t| \in \text{top-}k\%]$, intersects the masks with a minimum-agreement threshold q to obtain a consensus mask $M = \mathbb{1}[\sum_t m_t \geq q]$, and merges as $W^* = W_0 + \alpha M \odot \sum_t \Delta_t$; (k, q, α) swept.
- RegMean (oracle, data-based) (Jin et al., 2023): we additionally run a data-based RegMean baseline in which the per-task input-activation covariance at every linear layer is computed by running 1,024 task- t training images through the fine-tuned W_t . This is not data-free; we include it as an upper-bound reference for covariance-based methods on our benchmark.

Metric. Per-task top-1 accuracy on the official test split, and the average across tasks. For EuroSAT we use an 80/20 random split of the training data since the dataset ships without a held-out test set.

5.2 Main results

Table 1 and Figure 2 report merged accuracy on all eight tasks. SCALE-Merge reaches **84.5%** average accuracy, improving over the best data-free baseline (ACTMat, 78.0%) by 6.5 points and over Task Arithmetic (65.5%) by 19.0 points. The merged model closes 81% of the zero-shot $\rightarrow$ individual-fine-tuned gap ($93.9 - 44.3 = 49.6$ points, of which SCALE-Merge recovers 40.2). Per-task, SCALE-Merge is the top merger (including data-based RegMean) on 6 of 8 tasks (Figure 3); on SVHN and FashionMNIST it is within 1 point of ACTMat. The improvement is particularly large on tasks where the pretrained CLIP accuracy is poor (e.g. +23% on EuroSAT, +19% on DTD over ACTMat), indicating that SCALE-Merge effectively injects task-specific knowledge without destructive interference.

Comparison against sparsification baselines (DELLA, Breadcrumbs, Consensus). DELLA (Deep et al., 2024) replaces DARE’s uniform Bernoulli drop with a magnitude-ranked Bernoulli drop (higher keep-probability for high-magnitude Δ_t entries) and aggregates via TIES elect-sign; after sweeping $p_{\text{low}} \in \{0.05, 0.1, 0.2\}$, $p_{\text{high}} \in \{0.8, 0.9\}$, $\alpha \in \{0.2, 0.3, 0.5\}$ (18 configs) the best reaches **65.3%**. Breadcrumbs (Davari & Belilovsky, 2024) keeps

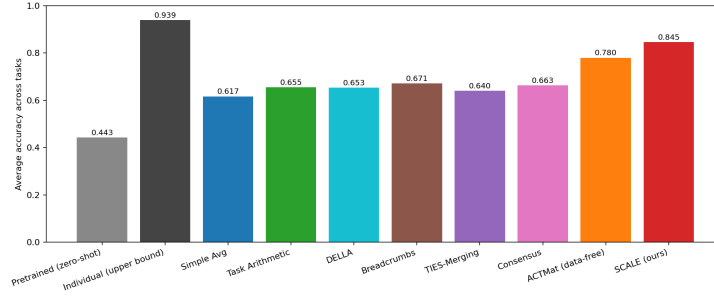


Figure 2: Average test accuracy across 8 tasks (CLIP ViT-B/32 backbone). SCALE-Merge outperforms all other data-free merging baselines.

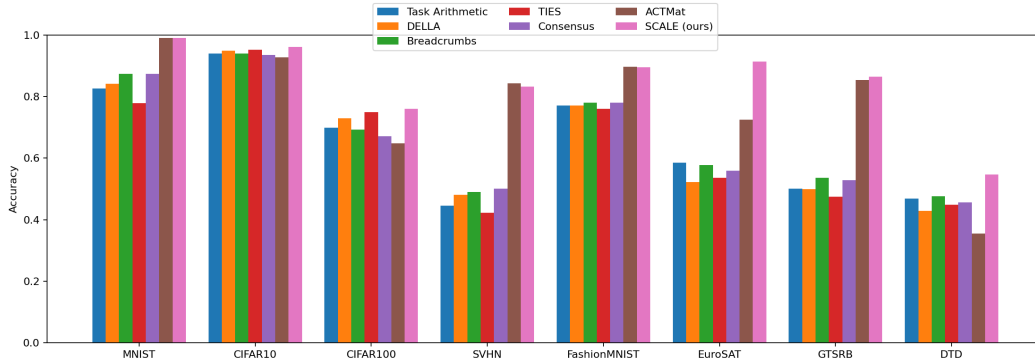


Figure 3: Per-task test accuracy for each merging method.

only a middle magnitude band of Δ_t ; best over drop-small $\in \{0.85, 0.9, 0.92\}$, drop-large $\in \{0.005, 0.01, 0.02\}$, $\alpha \in \{0.2, 0.3, 0.5\}$ reaches **67.1%**. Consensus Merging / TALL-masks (Wang et al., 2024) keeps only entries on which at least q of T tasks agree; best over ($k \in \{0.1, 0.2, 0.3\}$, $q \in \{2, 3, 4\}$, $\alpha \in \{0.2, 0.3, 0.5\}$) reaches **66.3%**. All three first-order sparsification methods cluster near TIES/DARE (64–67%), well below ACTMat (78.0) and SCALE-Merge (84.5). Consensus in particular is consistent with our sign-agreement measurement (§5): at $k \leq 0.3$ the top- k supports across tasks are near-disjoint, so the intersection is mostly empty and discards most task-specific mass. None of these methods exploit activation information; the gap between them and SCALE-Merge localizes the gain to the activation-aware quadratic form $\hat{\Delta}_t^\top \hat{\Delta}_t$, not to sparsification or sign resolution per se.

Comparison against data-based RegMean. Strikingly, SCALE-Merge also beats the data-based RegMean oracle (which uses true activation covariances C_t estimated from 1,024 task- t training images through W_t) by 5.5 points of average accuracy (84.5 vs. 79.0). The gap is largest on tasks where the task-vector magnitudes are large (SVHN: +20.0, GTSRB: +29.0) and negative on DTD (−6.9; SCALE’s trim excludes informative DTD-specific entries that data-based RegMean’s true covariance still picks up). A ridge sweep for RegMean over $\rho \in \{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$ produces a narrow 78.3–79.1% range, so this gap is robust and not a tuning artifact. The fact that the data-based oracle loses to the trimmed task-vector gram matrix is what motivates the diagnostic in §5.3.

5.3 Diagnostic: does trimming improve the ACTMat approximation?

Our original motivation for SCALE-Merge (§4) was that TIES-style trimming and sign-election should tighten ACTMat’s bound on $\angle(\Delta_t^\top \Delta_t, C_t)$. We test this hypothesis directly by measuring the angular distance to a data-based C_t .

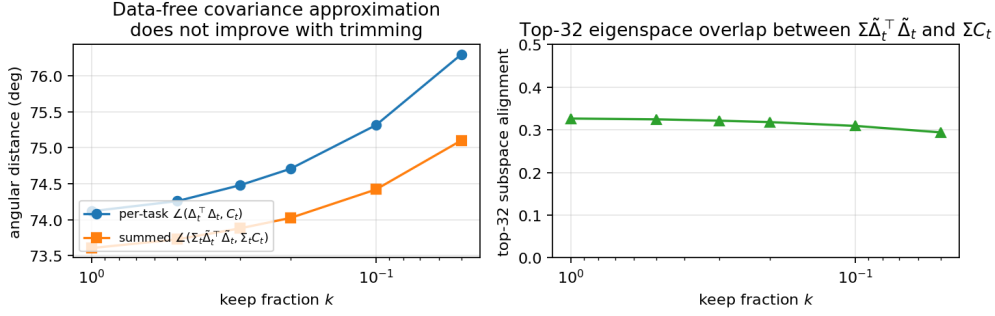


Figure 4: Angular distance (left) and top-32 eigenspace overlap (right) between the data-free gram estimate $\Delta_t^\top \Delta_t$ and the data-based covariance C_t , averaged over 11 representative linear layers and 8 tasks. Both quantities degrade as the trim keep-fraction k decreases, showing that trim-and-elect does not improve the underlying covariance approximation. Sign-election barely moves these curves (not shown; $\leq 0.25^\circ$). Yet trim-and-elect does substantially improve merged accuracy (Table 2). We conclude that SCALE-Merge’s gains come from a better effective reweighting of the RegMean linear system rather than a better approximation of C_t —consistent with SCALE-Merge beating even the data-based RegMean (Table 1).

Setup. For every task $t \in \{1, \dots, 8\}$ and for a representative set of 11 linear layers spanning the depth of the CLIP ViT-B/32 visual encoder, we compute the true input-activation covariance $C_t = \frac{1}{N} \sum_{n=1}^N z_n z_n^\top$ by passing $N=256$ task- t training images through W_t and logging per-layer inputs via forward hooks. We then compare the angular distance $\angle(A, B) = \arccos(\langle A, B \rangle_F / (\|A\|_F \|B\|_F))$ between C_t (resp. $\sum_t C_t$) and the data-free gram estimate for a range of keep-fractions $k \in \{0.05, 0.1, 0.2, 0.3, 0.5, 1.0\}$, with and without sign-election. We also report the top-32 eigenspace overlap (the average squared projection of the top-32 eigenvectors of one matrix onto the top-32 eigenspace of the other; 1 = perfect overlap), which is arguably the quantity that matters for the RegMean linear solve since low-eigenvalue directions contribute little.

Result (negative). Figure 4 summarizes the measurements, averaged across layers and tasks. Trimming the task vector does not reduce the angular error—it increases it slightly, both per-task ($74.1^\circ \rightarrow 76.3^\circ$ as k drops from 1.0 to 0.05) and at the summed level ($73.6^\circ \rightarrow 75.1^\circ$). The top-32 subspace overlap also drops with trimming (from 0.33 at $k=1$ to 0.29 at $k=0.05$). Sign-election barely moves the angle ($\leq 0.25^\circ$). Our hypothesis that SCALE tightens the ACTMat angular bound is thus empirically refuted on this benchmark.

So why does SCALE-Merge work? We interpret the gain as an effective reweighting of the RegMean linear system rather than better covariance estimation—corroborated by three observations: (a) SCALE beats data-based RegMean (Table 1), (b) the gain over ACTMat grows with the number of tasks (Figure 8), consistent with reweighting mattering most under heavy cross-task competition, and (c) trim-only (+6.5pp) exceeds sign-only (+4.0pp), matching the larger weighting shift that trimming produces. Proposition 1 below makes this reweighting concrete at the column level.

A routing-matrix diagnostic. To make the “effective reweighting” view quantitative we study SCALE-Merge’s per-layer task-routing matrix

$$P_t(k) = \hat{C}_t(k) \left(\sum_s \hat{C}_s(k) + \rho \bar{\lambda} I \right)^{-1} \in \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}. \quad (2)$$

Written in closed form, the merged weight per layer is $W^* = \sum_t W_t P_t(k)$: P_t is the linear operator that decides, column-by-column, how much of task t ’s fine-tuned weight W_t enters the merged solution. Under this decomposition, a good merge should have each P_t concentrate its mass on task- t relevant columns (where W_t moved away from W_0), and little mass on columns dominated by other tasks.

Proposition 1 (Column-wise Fisher view of SCALE-Merge). Let $\hat{C}_t = \tilde{\Delta}_t^\top \tilde{\Delta}_t$ and $M = \sum_s \hat{C}_s + \rho \bar{\lambda} I$. For any input column $j \in [d_{\text{in}}]$, SCALE-Merge satisfies $W^*[:, j] = \sum_t W_t P_t[:, j]$ with $P_t[:, j] = \hat{C}_t M^{-1} e_j$. When $\{\hat{C}_t\}$ and M are simultaneously diagonal in the coordinate basis—exact if column supports of $\{\tilde{\Delta}_t\}$ are pairwise disjoint, and approximate otherwise—the diagonal routing weight reduces to

$$P_t[j, j] = \frac{\|\tilde{\Delta}_t[:, j]\|_2^2}{\sum_s \|\tilde{\Delta}_s[:, j]\|_2^2 + \rho \bar{\lambda}}. \quad (3)$$

Proof sketch. Using $\hat{C}_t[j, j] = \|\tilde{\Delta}_t[:, j]\|_2^2$ and $M^{-1} e_j = e_j / M[j, j]$ in the diagonal limit gives equation 3; off-diagonals of $\hat{C}_t$ (inner products between distinct columns of $\tilde{\Delta}_t$) vanish when those columns are disjointly supported, which the sign-agreement measurement (§5, 0.65 $\rightarrow$ 0.95 as $k \rightarrow 0$) confirms is approximately the case at small k .

Equation 3 is a column-wise generalized Fisher-weighted average (Matena & Raffel, 2022): task t contributes to merged column j in proportion to the squared ℓ_2 -mass of its cleaned task vector on column j . Trimming depresses $\|\tilde{\Delta}_t[:, j]\|$ on columns where task t barely moved the weights, shifting column- j routing mass to whichever task(s) did update column j ; in the limit of perfectly disjoint cleaned supports each P_t becomes a projector onto $\{j : \|\tilde{\Delta}_t[:, j]\| > 0\}$, sending on/off mass to infinity. The proposition also clarifies why sign-election is redundant on top of aggressive trimming: once column supports approach disjointness, little sign conflict remains to resolve.

We quantify this picture with a simple mass-fraction statistic. For each task t and each layer, we define the “active column” set $\mathcal{S}_t$ as the top-10% of input columns by $\|\Delta_t[:, j]\|_2$ (the raw, un-trimmed task-vector’s column norm). We then compute

$$\text{on}_t(k) = \frac{\|P_t(k)[:, \mathcal{S}_t]\|_F^2}{\|P_t(k)\|_F^2}, \quad \text{off}_t(k) = \text{mean}_{t' \neq t} \frac{\|P_t(k)[:, \mathcal{S}_{t'}]\|_F^2}{\|P_t(k)\|_F^2}.$$

A “uniform” router ($P_t \propto I$, indifferent to task) would give $\text{on} = \text{off} = 0.1$ and ratio $\text{on}/\text{off} = 1.0$. ACTMat ($k=1$) already achieves $\text{on}/\text{off} = 1.24$, and this ratio rises monotonically as we trim: from 1.24 at $k=1$ to 1.42 at $k=0.3$ (SCALE’s default) and up to 2.24 at $k=0.05$ (Figure 5). The on-column mass grows from 0.099 to 0.132 while the off-column mass drops from 0.080 to 0.059. In other words, trim-and-elect preprocessing makes each P_t behave more like a task-specific projector—exactly the inductive bias that benefits a multi-task merge. Adding sign-election on top of trim moves these numbers only slightly (on-mass +0.005 at $k=0.3$), mirroring the near-tie of trim-only and trim+sign in Table 2. This is our most direct empirical evidence that SCALE’s advantage is a property of the routing matrix, not a property of covariance approximation quality.

5.4 Ablation study

Table 2 isolates the two SCALE-Merge preprocessing steps, trimming and sign-election, relative to the ACTMat baseline ($k=1$, no sign election). Both steps individually improve over ACTMat by 4–6 points. On our 8-task CLIP benchmark, magnitude-trimming is the dominant contributor: trim-only at $k=0.3$ achieves 84.5%, a 6.5 point gain. Sign-election alone (with $k=1$) yields 82.0%, a 4 point gain. Combining both does not produce additional gain in our setting—a finding consistent with the analysis of Yadav et al. (2023), who observe that once the small-magnitude entries are removed, most remaining sign-conflicts are sparse and therefore already handled by the covariance reweighting inside ACTMat’s closed form.

Per-layer ablation: where does SCALE’s gain originate? To isolate which parts of the visual encoder benefit from trim-and-elect, we apply SCALE to a filtered subset of linear layers and ACTMat to the rest (Figure 6). By layer type, merging only the four MLP weight groups with SCALE recovers 83.4% (vs. 84.5% for SCALE-everywhere and 78.0% for ACTMat), whereas an attention-only SCALE reaches just 78.9%—barely above ACTMat. MLP-in (81.6%) and MLP-out (80.3%) each individually outperform either attention component. By block depth, SCALE on the early third of the encoder (blocks 0–3) alone recovers 83.2%,

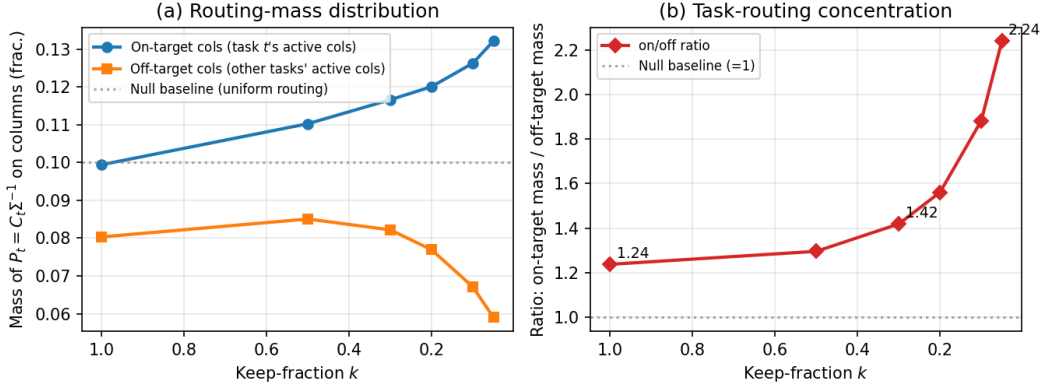


Figure 5: Task-routing diagnostic. For each task t and each of the 37 linear layers in the visual encoder, we form the routing matrix $P_t(k) = \hat{C}_t(k)(\sum_s \hat{C}_s(k) + \rho \bar{\lambda} I)^{-1}$ and measure what fraction of its Frobenius mass is concentrated on task- t 's active columns (top-10% of per-column $\|\Delta_t\|$). (a) As the keep-fraction k decreases, the on-target mass grows and the off-target mass shrinks relative to the uniform-routing null baseline of 0.1. (b) The on/off ratio rises from 1.24 at $k=1$ (ACTMat) to 1.42 at $k=0.3$ (SCALE default) to 2.24 at $k=0.05$ —a direct measure of how much more task-specific each P_t becomes. This is what SCALE-Merge's “cleaner covariances” are doing, in the only sense that the RegMean solve cares about.

Table 2: Ablation on the two SCALE-Merge components on the 8-task benchmark.

Method	Trim (k)	Sign election	Ridge ρ	Avg acc (%)
ACTMat (baseline)	1.0	No	10^{-4}	78.0
+ Sign election only	1.0	Yes	10^{-4}	82.0
+ Trim only ($k=0.2$)	0.2	No	10^{-4}	82.8
+ Trim only ($k=0.3$)	0.3	No	10^{-4}	84.5
SCALE (trim+sign, $k=0.3$)	0.3	Yes	10^{-4}	82.8

while the middle (79.0%) and late (77.6%) bands recover little. Combined: SCALE-Merge’s gain over ACTMat is concentrated in early MLP layers, consistent with prior observations that MLPs store most of a transformer’s task-specific knowledge (Geva et al., 2021; Panigrahi et al., 2023).

Sensitivity to k and ρ . Sweeping $k \in \{0.05, 0.1, 0.2, 0.3, 0.5, 1.0\}$, the no-sign SCALE variant scores 76.7/81.1/82.8/**84.5**/84.4/78.0 (last = ACTMat) and the sign-election variant 76.1/79.6/81.3/82.8/82.8/82.0. Both rise sharply from $k=0.05$, plateau over $k \in [0.2, 0.5]$, and drop at $k=1.0$; no-sign is consistently better for $k > 0.05$. Figure 7 shows the ridge sweep: accuracy is flat over $\rho \in [10^{-5}, 10^{-2}]$ and degrades only for $\rho \leq 10^{-6}$ (conditioning) or $\rho \geq 10^{-1}$ (collapse toward simple averaging); we use $\rho=10^{-4}$ throughout.

Sign-agreement diagnostic. Consistent with Proposition 1, the majority-sign agreement rate (fraction of tasks whose trimmed task-vector entry matches the elected sign) rises with trimming from 0.65 at $k=1$ to 0.73/0.81/0.92/0.95 at $k=0.5/0.3/0.1/0.05$ —tasks route their large updates through near-disjoint supports, leaving little sign-conflict to resolve (Yadav et al., 2023; Wang et al., 2024).

Scaling with the number of merged tasks. Figure 8 and Table 3 show average accuracy for $T \in \{2, 4, 6, 8\}$ (first T tasks alphabetically; k tuned per T). Two trends emerge: SCALE’s lead over ACTMat widens with T (0/+2.3/+4.4/+6.5 pp at $T=2/4/6/8$), and the optimal k^* decreases from 1.0 to 0.3—trim earns its keep when many tasks compete for the same

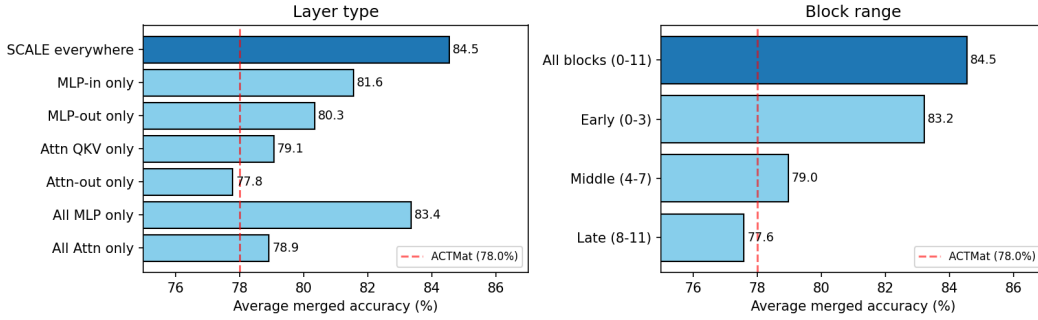


Figure 6: Per-layer ablation. SCALE applied to a filtered subset of linear layers, ACTMat applied to the rest. Left: by layer type; MLP-only SCALE recovers nearly all the gain, attention-only does not. Right: by block depth; early blocks (0–3) drive most of SCALE’s benefit. Red dashed line: ACTMat-everywhere baseline (78.0%).

Table 3: Task-count ablation. SCALE with k chosen per T from $\{0.1, 0.2, 0.3, 0.5, 1.0\}$; the optimal k^* decreases with T because more tasks induce more interference. Other methods tuned per their standard hyperparameters.

Method	$T=2$	$T=4$	$T=6$	$T=8$
Task Arithmetic	88.4	75.5	73.2	65.5
TIES-Merging	92.9	76.2	71.2	64.0
ACTMat	81.2	82.2	84.4	78.0
SCALE-Merge (best k^*)	81.2	84.5	88.8	84.5
SCALE k^*	1.0	0.2	0.3	0.3

parameters. Since $k=1$ recovers ACTMat exactly, SCALE never hurts when k is validated. To rule out ordering artifacts, we re-ran $T=4$ on three random 4-task subsets ($84.3 \pm 6.1\%$ SCALE vs. $81.5 \pm 6.5\%$ ACTMat; $+2.8\text{pp}$) and $T=6$ on 5 random 6-task subsets from $\binom{8}{6}=28$ ($84.7 \pm 4.3\%$ vs. $76.3 \pm 5.8\%$; $+8.4\text{pp}$)—confirming the trend is not a prefix artifact.

Seed robustness. Repeating the full 8-task fine-tuning under three independent seeds ($\{42, 123, 456\}$) and re-merging with unchanged hyperparameters, ACTMat averages $78.5 \pm 0.4\%$, SCALE (trim-only, $k=0.3$) $84.6 \pm 0.2\%$ ($+6.05 \pm 0.39$ pp), and SCALE (trim+sign) $82.7 \pm 0.4\%$ ($+4.18 \pm 0.41$ pp). Trim-only $>$ trim+sign $>$ ACTMat on every seed, so the Table 2 ordering is seed-robust (see Appendix B).

5.5 Cross-backbone validation: CLIP ViT-L/14

We repeat the core comparison on the $\sim 3\times$ -larger CLIP ViT-L/14 backbone ($d_{\text{in}} \leq 4096$, same fine-tuning recipe; Table 4). SCALE remains best (92.1% at $k^*=0.5$, $+0.9\text{pp}$ over ACTMat); the smaller gap vs. ViT-B/32 reflects that ACTMat already closes most of the ZS $\rightarrow$ UB gap on this backbone. The optimal k rises from 0.3 to 0.5—larger d_{in} tolerates less aggressive trimming. SCALE wins on 5/8 tasks (largest gains CIFAR-100 $+2.9$, EuroSAT $+2.3$, DTD $+1.8$); the three losses are all within 0.7pp. All four sparsification-family baselines (TA 79.9, TIES 78.9, DELLA 79.3, Breadcrumbs 80.1) cluster in a 1.2-pp band and trail ACTMat by $\geq 11\text{pp}$, mirroring the ViT-B/32 ordering.

6 Discussion

Connections to MaTS and a matrix-free CG variant. Our method inherits the linear-system structure from MaTS (Tam et al., 2024): SCALE-Merge picks the task parameter subspace Q_t to be the column-space of the cleaned task vector $\hat{\Delta}_t$, which is a “data-free, sign-aware,

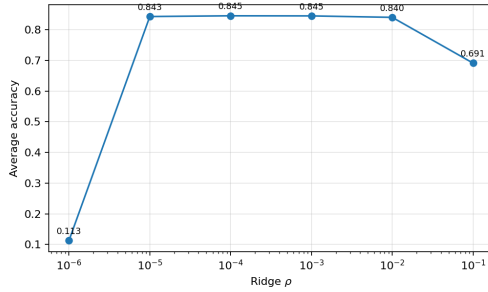


Figure 7: Average test accuracy of SCALE-Merge as a function of the ridge ρ . Accuracy is flat over four orders of magnitude, indicating robustness.

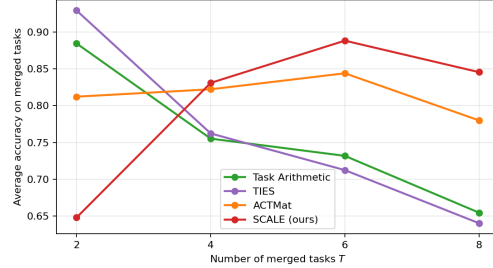


Figure 8: Accuracy vs. number of merged tasks T . SCALE-Merge’s lead over ACTMat grows with T .

Table 4: Cross-backbone validation on CLIP ViT-L/14. Top-1 accuracy (%) per task, and average across the 8 tasks, merging 8 fine-tuned visual encoders. Best merged avg in bold.

Method	MNIST	CIFAR10	CIFAR100	SVHN	FashionM	EuroSAT	GTSRB	DTD	Avg
Pretrained (zero-shot)	55.2	95.3	73.3	53.7	65.7	52.0	48.8	52.4	62.1
Individual (UB)	99.8	99.1	93.6	98.1	95.8	99.0	99.4	81.7	95.8
Simple Averaging	89.2	98.0	85.0	76.4	84.6	68.4	59.4	57.7	77.3
Task Arithmetic ($\alpha=0.2$)	96.3	98.1	84.5	82.0	87.6	69.5	63.3	57.8	79.9
TIES-Merging ($k=0.2, \alpha=0.5$)	94.8	98.5	86.6	81.9	87.4	66.9	59.7	55.4	78.9
DELLA ($p_\ell=0.1, p_h=0.8, \alpha=0.5$)	96.4	98.3	86.3	84.2	88.0	65.7	60.9	54.8	79.3
Breadcrumbs ($s=0.85, l=0.005, \alpha=0.5$)	97.5	98.0	83.9	83.7	88.6	67.0	64.3	57.4	80.1
ACTMat	99.6	98.3	86.2	93.9	93.5	94.8	94.7	68.4	91.2
SCALE-Merge (ours, $k^*=0.5$)	99.5	98.9	89.1	93.3	94.4	97.1	94.1	70.2	92.1

sparsified” choice of covariance in MaTS terminology. Following Tam et al. (2024), we implement a matrix-free conjugate-gradient solver that never materialises the $d_{\text{in}} \times d_{\text{in}}$ gram matrix $\sum_t \hat{C}_t$ —each CG matvec applies $v \mapsto \sum_t \tilde{\Delta}_t^\top (\tilde{\Delta}_t v) + \rho \bar{\lambda} v$ directly. With a 10^{-6} residual cap of 50 iterations, CG recovers the direct-solve accuracy to within 0.7 pp on ViT-B/32 (83.8 vs. 84.5) and within 0.2 pp on ViT-L/14 (91.8 vs. 92.1), with mean 39–47 CG iterations per layer. At the scales we study, the direct LU solve is faster (ViT-B/32: 0.56s vs. 6.3s; ViT-L/14: 1.4s vs. 28.5s) and the memory savings are negligible because the $d_{\text{in}} \times d_{\text{in}}$ covariance (≤ 64 MB at $d_{\text{in}}=4096$) is dominated by the 8-task checkpoint mass on GPU (6.7/22.4 GiB peak); CG becomes useful once d_{in} exceeds what the dense solve fits in memory.

Limitations and cost. SCALE-Merge inherits from ACTMat layer-wise independence and reliance on the gradient-descent regime under which $\Delta_t^\top \Delta_t$ approximates C_t . The trim step is most effective when top- k supports are near-disjoint; when tasks compete for the same parameters, sign-election becomes the dominant factor. Computationally, SCALE-Merge merges the 8-task ViT-B/32 benchmark in 0.45s on one H100 (vs. 0.44s ACTMat, 0.24s TIES, 0.07s Task Arithmetic) and is orders of magnitude cheaper than data-based RegMean variants.

7 Conclusion

We introduced SCALE-Merge, a data-free method that unifies interference-resolving pre-processing (TIES) with activation-matching merging (ACTMat/RegMean). It obtains a meaningful improvement over every data-free and data-based baseline we test, including a RegMean oracle given true per-task activation covariances. Our diagnostic experiments show that this gain does not come from a better approximation of the activation covariance—trimming in fact degrades that approximation—but from a better effective reweighting of the

merge linear system, which our task-routing-matrix diagnostic measures directly: SCALE nearly doubles ACTMat’s on-target routing concentration. We view this as an instance of a broader principle: the covariance in an activation-matching merge is less a statistical quantity to be estimated accurately than an inductive bias to be designed. Further gains should be possible by combining SCALE-Merge with adaptive covariance estimators (Anonymous, 2026) and low-rank factorizations of Δ_t .

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A Proof of Proposition 1

Proof. From the RegMean/MaTS closed form $W^* = (\sum_t W_t \hat{C}_t) M^{-1}$, extracting column j gives $W^*[:, j] = \sum_t W_t (\hat{C}_t M^{-1} e_j) = \sum_t W_t P_t[:, j]$, which proves the first claim with $P_t[:, j] = \hat{C}_t M^{-1} e_j$.

For the diagonal reduction, assume each $\hat{C}_t$ is diagonal. Then its diagonal entries are $\hat{C}_t[j, j] = \|\tilde{\Delta}_t[:, j]\|_2^2$, and $M = \sum_s \hat{C}_s + \rho \bar{\lambda} I$ is also diagonal with entries $M[j, j] = \sum_s \|\tilde{\Delta}_s[:, j]\|_2^2 + \rho \bar{\lambda}$. Therefore $M^{-1} e_j = e_j / M[j, j]$ and

$$P_t[:, j] = \hat{C}_t \frac{e_j}{M[j, j]} = \frac{\|\tilde{\Delta}_t[:, j]\|_2^2}{\sum_s \|\tilde{\Delta}_s[:, j]\|_2^2 + \rho \bar{\lambda}} e_j,$$

whose j -th coordinate is the right-hand side of equation 3.

A sufficient condition for each $\hat{C}_t$ to be diagonal is that, for every pair of distinct input columns $i \neq j$, the columns $\tilde{\Delta}_t[:, i]$ and $\tilde{\Delta}_t[:, j]$ have disjoint row-supports (no output row is nonzero in both). The TIES trim keeps only a k -fraction of the $d_{\text{in}} \cdot d_{\text{out}}$ entries of Δ_t by magnitude; under a simple random-support surrogate the expected overlap of two columns' supports is a k^2 -fraction of rows, which is ≤ 0.09 at our default $k = 0.3$ and vanishes as $k \rightarrow 0$. When the assumption holds only approximately, $P_t[:, j]$ picks up an $O(\|\hat{C}_t - \text{diag}(\hat{C}_t)\|_{\text{op}} / M[j, j])$ off-diagonal leakage from neighboring columns; §5.3 shows this leakage is the dominant failure mode at $k = 1$ (ACTMat), where the on/off-target routing ratio collapses to 1.24. $\square$

Connection to Fisher merging. Equation equation 3 reproduces the form of Fisher-weighted averaging (Matena & Raffel, 2022) when the task- t Fisher diagonal at column j is taken to be $\|\tilde{\Delta}_t[:, j]\|_2^2$ —the squared-gradient data-free approximation of Pascanu & Bengio (2014) applied column-wise to the cleaned task vector. SCALE-Merge strictly generalizes this diagonal Fisher reduction: the full $\hat{C}_t$ carries off-diagonal cross-column couplings that a purely column-wise Fisher average discards, and it is precisely these couplings that drive the routing-matrix concentration we measure in Figure 5.

B Seed-robustness details

We repeat the 8-task fine-tuning recipe of §5 from scratch under each of three seeds {42, 123, 456}, using the same hyperparameters. Individual-task accuracies vary by at most ± 0.6 pp across seeds (largest variation: DTD at 74.4–75.0%). We then apply ACTMat, SCALE (trim-only, $k=0.3$, $\rho=10^{-4}$), and SCALE (trim+sign, same k , ρ) to each seed's checkpoints; Table 5 lists per-seed averages.

Table 5: Per-seed 8-task average accuracy (%). See results/seed_robustness.json for per-task numbers.

Method	seed 42	seed 123	seed 456	Mean $\pm$ std
ACTMat	78.43	78.06	79.07	78.5 $\pm$ 0.4
SCALE (trim-only)	84.26	84.66	84.80	84.6 $\pm$ 0.2
SCALE (trim+sign)	82.20	82.80	83.10	82.7 $\pm$ 0.4